Chip-to-Radiator Heat Budget for Orbital AI Compute

Oct 31, 2026 · @Prachet Poddar

Summary

For a 700 W accelerator whose die face is held at 60 °C, a solder interface and a bonded joint together cost about 8 K. Spreading the heat from the die out to the radiator joint costs 25 to 83 K on 5 mm aluminium, copper or diamond-class plates. A placeholder hot spot, a tenth of the die carrying half the power, puts the peak 36 to 75 K above the die's mean, depending on the spreader.

Near 60 °C, every kelvin lost before the radiator needs about 1.2% more radiator area. With a solder interface in place, the remaining levers are the spreader's conductance times thickness and the die's power map. A polymer interface is the exception: at 1 W/m/K it can cost as much as spreading does.

The model is first order: one chip on an isothermal panel. The conduction totals are lower bounds for the stated materials and assumed geometry, while the hot-spot peaks rest on placeholder power maps and could be higher or lower. Three Python scripts produce every figure from public datasheets, patents and stated assumptions. They describe no company's actual hardware. The GAO's April 2026 spotlight calls cooling at data-centre scale in space unproven, and this note asks where along the conduction path the temperature budget goes.

Why each kelvin costs radiator area

A radiator held 1 K cooler needs 1.21% more area near 60 °C, and a 10 K drop costs 12.96%. Radiated power per unit area grows as the fourth power of temperature, so any drop between the die and the radiator forces a cooler, larger radiator.

$A_{rad} = Q / [\, n \cdot \eta \cdot (\varepsilon \cdot \sigma \cdot T_{rad}^4 - q_{abs}) \,]$, and to first order $dA / A \approx -4 \cdot dT / T_{rad}$.

Here Q is the chip power, n the number of radiating sides, η the panel efficiency, ε the emissivity and q_{abs} the absorbed environmental flux. With ε = 0.85, η = 0.95 and an ideal deep-space sink, one 700 W chip needs these radiator areas with no conduction loss at all:

Radiator temperature	One-sided area (m²)	Two-sided area (m²)
60 °C	1.241	0.621
80 °C	0.983	0.491

IEEE Spectrum (Cavalier, 11 June 2026) gives 1.4 m² for the same 700 W chip at 60 °C against a 3 K background. The gap is consistent with different emissivity and efficiency assumptions, which the parts of the article read here do not state. Every result below is relative to this zero-loss baseline, which no real stack reaches.

Stage 1: the interfaces

A solder interface plus a 400 cm² bonded joint costs 7.70 K, a 9.8% area penalty with the die face held at 60 °C. A polymer interface can cost far more. Heat leaves the die at 8.6×10^5 W/m² (700 W over 814 mm²), so bond-line resistance matters.

Layer	Temperature rise (K)	Basis
Solder TIM, 52In/48Sn	3.35	0.039 cm²·°C/W at 100 psi, 4 mil bond line (Indium vendor value)
Solder TIM, pure In	4.42	0.0514 cm²·°C/W, same brochure
Polymer TIM, k = 5 W/m/K, 25 / 50 / 100 µm	4.30 / 8.60 / 17.20	5 W/m/K is the upper bound for non-metallic TIMs in US 7663242; thicknesses assumed
Polymer TIM, k = 1 W/m/K, 25 / 50 / 100 µm	21.50 / 43.00 / 86.00	Same patent: typically below 1 W/m/K
Bonded heat-pipe joint, 400 cm²	4.34	2.6 W/in²·°C design value, US 12515822; footprint assumed
Bonded heat-pipe joint, 900 cm²	1.93	Same value

The rise is the die-face temperature increase at a fixed radiator temperature. The joint value's reference area in the patent is not confirmed, so the joint rows could be off by that ratio.

Absorbed environmental flux makes every kelvin dearer. At 100 and 200 W/m², the one-sided baseline grows from 1.241 m² to 1.492 and 1.871 m², and the 7.70 K stack's penalty grows from 9.8% to 12.0% and 15.6%.

Stage 2: spreading from the die to the joint

Spreading is the largest term with a solder interface: 25 to 83 K for 5 mm plates, against 7.70 K for the solder interface and joint together. Heat enters through 8.14 cm² but the joint footprint is 400 cm², a 49× spread. The model is a uniform-flux disk on a circular plate with the bonded joint on its bottom face.

Plate, 5 mm, 400 cm² footprint	TIM (K)	Plate 1D (K)	Spreading (K)	Joint (K)	Total (K)	Radiator area vs zero loss
Al 6061	3.35	0.52	83.45	4.34	91.67	3.62×
Cu	3.35	0.22	49.14	4.34	57.06	2.12×
Diamond-class (k = 1000 W/m/K, assumed)	3.35	0.09	24.98	4.34	32.76	1.51×

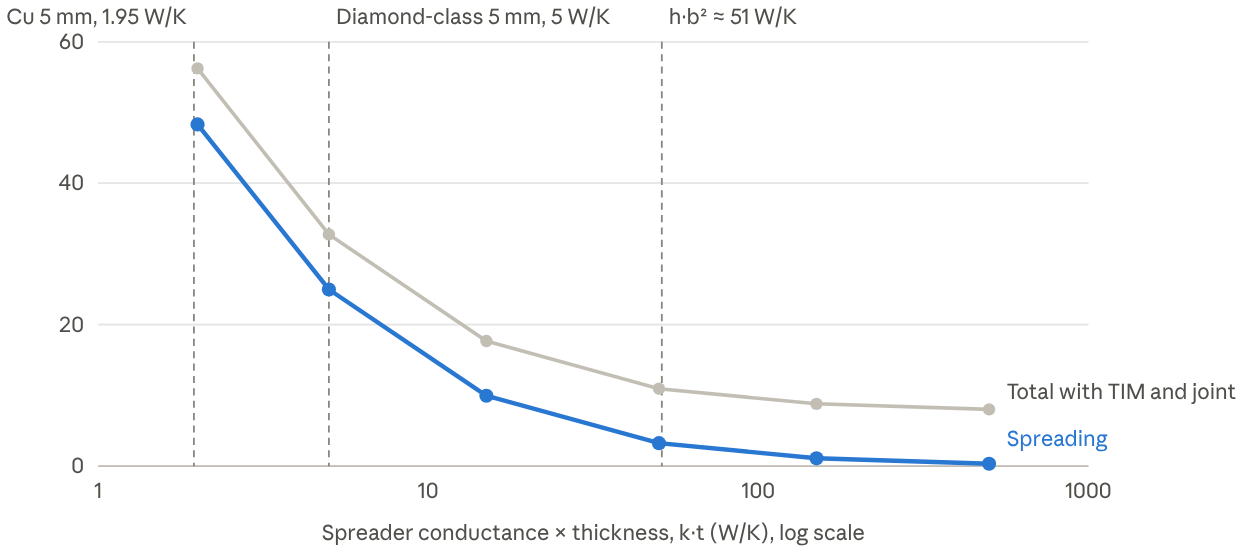
The area multiple holds the mean die face at 60 °C with no absorbed flux. Multiples do not add across layers, because of the fourth-power law.

A bigger footprint does not help. For Cu 5 mm the total is 58.82, 57.25, 57.06 and 57.05 K at 100, 200, 400 and 900 cm²: a larger joint cuts the joint drop but pushes the heat farther, and the two cancel.

Conductance times thickness does help. The joint only draws heat out to a fin length of $\sqrt{(k \cdot t/h)}$, 22 mm for Cu 5 mm against a 113 mm footprint radius. For the whole footprint to work, $k \cdot t$ must approach $h \cdot b^2$, about 51 W/K. Cu 5 mm gives 1.95 W/K and diamond-class 5 mm gives 5 W/K.

Spreading drop reaches about 1 K only near k·t = 150 W/K

Temperature drop (K) for a 700 W die on a 400 cm² joint, 5 mm plate with isotropic effective conductivity



spreading_stage.py, Table D, run 31 October 2026

It is a knee, not a threshold. Reaching about 1 K of spreading takes k·t near 150 W/K, and the total then flattens toward the 7.70 K the interfaces cost. A real vapour chamber or heat-pipe panel is anisotropic, and no effective conductivity for one was taken from any source.

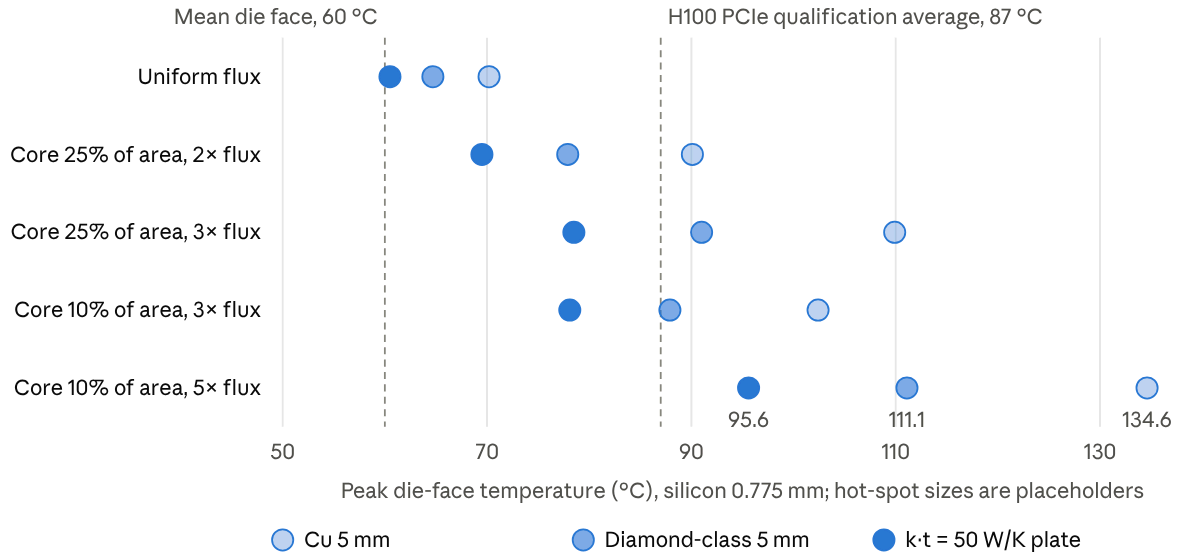
Stage 3: silicon and hot spots

Once the spreader is good, the hot spot sets the peak. With a k·t = 50 W/K plate, the mean die rise is 16 K, but a placeholder core holding half the power on a tenth of the area puts its centre 35.6 K above the mean. On Cu 5 mm the same core puts it 74.6 K above. Uniform heating alone gives 0.5 K and 10.2 K of peak above mean on those two plates.

The model adds a silicon die (k = 130 W/m/K) heated on the face away from the plate, a local die-attach resistance of 3.9×10^{-6} m²K/W (the 52In/48Sn solder value from Stage 1), and a two-zone heat map. With uniform flux, silicon adds 1.8 to 2.0 K at 0.3 mm thickness and 4.8 to 5.1 K at 0.775 mm. The hot-spot sizes and power shares are placeholders, because no public power map of any chip was used.

Peak die-face temperature, with the mean die face held at 60 °C

Hot-spot cases above the 87 °C reference, which is not a limit: Cu 5 mm 4 of 4, diamond-class 3 of 4, k·t = 50 W/K plate 1 of 4



hotspot_silicon.py, Table H2, run 31 October 2026

Two reference temperatures frame the chart, and neither is a junction limit. IEEE Spectrum calls 60 °C the usual sweet spot for GPU longevity. NVIDIA's H100 PCIe product brief qualifies the card at a GPU sensor average of 87 °C. That is a sensor average on a different card, not a peak junction limit for the SXM part, which the sources read here do not give.

Thin silicon matters more as the spreader improves. Going from 0.775 mm to 0.1 mm cuts the peak rise of the 10% core by 11% on Cu, 22% on diamond-class and 42% on the k·t = 50 W/K plate.

Silicon conducts less when hot, and the model uses one value. The Electronics Cooling fit is printed as k = 1.5 (T/300)^(-4/3) with no units. Read as 150 W/m/K, it gives about 130 W/m/K at 60 °C and 112 at 100 °C. The same article quotes another source at 145 W/m/K at 100 °C for pure silicon, so the value at the hot core is uncertain by about that much. Dropping k from 130 to 100 raises the 10% core's peak rise from 140.5 to 149.5 K on Cu, 90.8 to 98.9 K on diamond-class, and 51.9 to 59.7 K on the k·t = 50 W/K plate.

Questions for an orbital compute design

The model cannot answer these, and each one moves the result by more than the choice between solder interfaces does.

1. What is the effective k·t of the path from die to panel, and how anisotropic is it?

2. What does the bonded joint's conductance measure at die scale, and over which area is it quoted?
3. What is the peak-to-mean die temperature on the panel, under the chip's real power map?
4. What junction limit applies at the hot core, and how far does silicon conductivity fall there?

How the numbers were checked

The three scripts run 109 self-checks before printing any table, and all pass: 31 in the radiator script, 41 in the spreading script and 37 in the silicon and hot-spot script.

- **Two solvers for spreading.** The series solution matches an independent finite-volume solve within 0.5% in three deliberately unequal cases. The finite-volume solve conserves energy to 1 part in a million and, in one case, changes under 0.3% on a twice-refined grid.
- **An outside solver for hot spots.** Four stored reference values from a solver that a separate reviewer wrote from the problem statement alone are built in as anchors. The layered model matches all four, mean and peak, within 0.2%. That solver is not part of the release, so these values cannot be regenerated from it.
- **Known answers and limits.** Checks include the half-space spreading limit, a thin-plate closed form, hand-computed layer values and energy balance.
- **Checks that must fail.** Several checks are run against deliberately broken versions of the model, such as a T^3 law, a dropped efficiency or a joint conversion with the inch not squared, and must catch each one.

Verification stops at internal consistency and agreement on an idealised geometry. None of it says anything about a real, square, non-uniformly heated die on a shared panel, and table rows are not tested one by one, only the functions they use.

Limits of the model

- One chip on an isothermal panel, with no neighbouring dies.
- Uniform die flux unless a hot spot is stated, and then one centred hot spot with placeholder size and power share.

- No contact resistance beyond the die-attach value. Rows at and above k·t = 15 W/K use an isotropic effective conductivity, while a real vapour chamber or heat-pipe panel is anisotropic.
- One silicon conductivity, though silicon conducts less where it is hottest.
- The joint conductance is a patent design value with an unconfirmed reference area, and the panel efficiency of 0.95 is a vendor statement.
- The 814 mm² die area comes from a hardware database, not from NVIDIA.

Inputs and sources

The three scripts need only Python, numpy and scipy, and every input is tagged in the code as published or assumed. Code, run logs and licence (MIT): github.com/prachetpoddar/orbital-thermal-note.

Input	Value	Source
Chip power	700 W	NVIDIA H100 datasheet, Sep 2022: SXM "Up to 700W (configurable)"
Die area	814 mm²	TechPowerUp, H100 SXM5 (not an NVIDIA figure)
Reference die temperature	60 °C	IEEE Spectrum, 11 June 2026
GPU qualification temperature	87 °C sensor average	NVIDIA H100 PCIe product brief, Table 5
Copper	390 W/m/K	Copper Development Association, C10100: 226 Btu/(ft·h·°F) at 68 °F, about 391
Aluminium 6061-T6	167 W/m/K	nTop material sheet, citing MatWeb
Silicon	130 W/m/K at 60 °C	Electronics Cooling, May 1998 fit
Solder TIM	0.039 and 0.0514 cm²·°C/W	Indium Corporation TIMs brochure 99342 R4, at 0.004 in bond line under a "Thermal Resistance at 100psi" header (vendor values)

Input	Value	Source
Polymer TIM bounds	5 and 1 W/m/K	US 7663242: non-metallic TIMs "generally does not exceed about 5 W/mK", "typically less than about 1 W/mK"
Joint conductance	2.6 W/in²·°C	US 12515822 (Lockheed Martin): design value for a film-adhesive joint bonding embedded heat pipes to the panel; bonded area not given
Panel efficiency	0.95	Advanced Cooling Technologies: "Typical CCHP-embedded radiator designs can produce a radiator efficiency of over 95%" (vendor claim)
Emissivity	0.85	Assumed, typical of white coatings
Diamond-class plate	1000 W/m/K	Assumed round figure

Every linked source was opened on 31 October 2026. Context: GAO-26-109012, Data Centers in Space, April 2026.